

Programming & Data Structures (PDS)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 34

Question 1 of 34 | GATE 2025 Set 1 (CS1) | Q.34 (Computer Science & IT)

Q.34

```
#include <stdio.h>
void foo(int *p, int x){
    *p=x;
}
int main(){
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo(z,b);
    printf("%d",a);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 2 of 34 | GATE 2025 Set 1 (CS1) | Q.65 (Computer Science & IT)

Q.65

In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i -th probe in the open address table is $[h_1(k) + i h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 3 of 34 | GATE 2025 Set 2 (CS2) | Q.13 (Computer Science & IT)

Q.13

Consider a binary tree T in which every node has either zero or two children.

Let $n > 0$ be the number of nodes in T .

Which ONE of the following is the number of nodes in T that have exactly two children?

- (A) $n - 2$
- (B) $n - 1$
- (C) n

(D) $n + 12$

Question 4 of 34 | GATE 2025 Set 2 (CS2) | Q.19 (Computer Science & IT)

Q.19

Consider the following C program:

```
#include <stdio.h>
void stringcopy(char *, char *);
int main(){
char a[30] = "@#Hello World!";
stringcopy(a, a + 2);
printf("%s\n", a);
return 0;
}
void stringcopy(char *s, char *t) {
while(*t)
*s++ = *t++;
}
```

Which ONE of the following will be the output of the program?

- (A) @#Hello World!
- (B) Hello World!
- (C) ello World!
- (D) Hello World!d!

Question 5 of 34 | GATE 2025 Set 2 (CS2) | Q.33 (Computer Science & IT)

Q.33

```
int x=126,y=105;
do {
if(x>y) x=x-y;
else y=y-x;
} while(x!=y);
printf("%d",x);
```

The output of the given C code segment is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 6 of 34 | GATE 2025 Set 2 (CS2) | Q.38 (Computer Science & IT)

Q.38

A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures:

P: Unsorted doubly linked list with pointers to the head node and tail node

of the list.

Q: Min-heap implemented using an array.

R: Binary Search Tree.

Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?

- (A) P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n)$
- (B) P: $\Theta(1)$, Q: $\Theta(n \log n)$, R: $\Theta(n)$
- (C) P: $\Theta(n)$, Q: $\Theta(n \log n)$, R: $\Theta(n^2)$
- (D) P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n \log n)$

Question 7 of 34 | GATE 2025 Set 2 (CS2) | Q.62 (Computer Science & IT)

Q.62

Consider the following C program:

```
#include <stdio.h>
int main(){
    int a;
    int arr[5] = {30,50,10};
    int *ptr;
    ptr = &arr[0] + 1;
    a = *ptr;
    (*ptr)++;
    ptr++;
    printf("%d", a + (*ptr) + arr[1]);
    return 0;
}
```

The output of the above program is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 8 of 34 | GATE 2025 Set 2 (CS2) | Q.63 (Computer Science & IT)

Q.63

Consider the following C program:

```
#include <stdio.h>
int g(int n) {
    return (n+10);
}
int f(int n) {
    return g(n*2);
}
int main() {
    int sum, n;
```

```

sum=0;
for (n=1; n<3; n++)
sum += g(f(n));
printf ("%d", sum);
return 0;
}

```

The output of the given C program is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 9 of 34 | GATE 2023 Single Session | Q.11 (Computer Science & IT)

Q.11

Which one of the following sequences when stored in an array at locations $A[1], \dots, A[10]$ forms a max-heap?

- (A) 23, 17, 10, 6, 13, 14, 1, 5, 7, 12
- (B) 23, 17, 14, 7, 13, 10, 1, 5, 6, 12
- (C) 23, 17, 14, 6, 13, 10, 1, 5, 7, 15
- (D) 23, 14, 17, 1, 10, 13, 16, 12, 7, 5 CS

Question 10 of 34 | GATE 2023 Single Session | Q.12 (Computer Science & IT)

Q.12

Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list.

Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?

- (A) SLLdel is $O(1)$ and DLLdel is $O(n)$
- (B) Both SLLdel and DLLdel are $O(\log(n))$
- (C) Both SLLdel and DLLdel are $O(1)$
- (D) SLLdel is $O(n)$ and DLLdel is $O(1)$ CS

Question 11 of 34 | GATE 2023 Single Session | Q.46 (Computer Science & IT)

Q.46

Consider the C function foo and the binary tree shown.

```

typedef struct node {
int val;

```

```

struct node *left, *right;
} node;
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}

```

10
5
3
8
11
13

When foo is called with a pointer to the root node of the given binary tree, what will it print?

- (A) 3 8 5 13 11 10
- (B) 3 5 8 10 11 13
- (C) 3 8 16 13 24 50
- (D) 3 16 8 50 24 13 CS

Question 12 of 34 | GATE 2023 Single Session | Q.58 (Computer Science & IT)

Q.58

Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q , both of which are initially empty.

- I: push the elements of a from a_0 to a_5 in that order into S .
 - II: enqueue the elements of a from a_0 to a_5 in that order into Q .
 - III: pop an element from S .
 - IV: dequeue an element from Q .
 - V: pop an element from S .
 - VI: dequeue an element from Q .
 - VII: dequeue an element from Q and push the same element into S .
 - VIII: Repeat operation VII three times.
 - IX: pop an element from S .
 - X: pop an element from S .
- The top element of S after executing the above operations is

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 34 | GATE 2022 Single Session | Q.21 (Computer Science & IT)

Q.21

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
    int x = 1, z[2] = {10, 11};
    int *p = NULL;
    p = &x;
    *p = 10;
    p = &z[1];
    *(&z[0] + 1) += 3;
    printf("%d, %d, %d\n", x, z[0], z[1]);
    return 0;
}
```

- (A) 1, 10, 11
- (B) 1, 10, 14
- (C) 10, 14, 11
- (D) 10, 10, 14

Question 14 of 34 | GATE 2022 Single Session | Q.43 (Computer Science & IT)

Q.43

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
    int a[3][3][3] =
    {{1, 2, 3, 4, 5, 6, 7, 8, 9},
    {10, 11, 12, 13, 14, 15, 16, 17, 18},
    {19, 20, 21, 22, 23, 24, 25, 26, 27}};
    int i = 0, j = 0, k = 0;
    for( i = 0; i < 3; i++){
        for(k = 0; k < 3; k++){
            printf("%d ", a[i][j][k]);
        }
        printf("\n");
    }
}
```

```
return 0;
}
```

- (A) 1 2 3 10 11 12 19 20 21
- (B) 1 4 7 10 13 16 19 22 25
- (C) 1 2 3 4 5 6 7 8 9
- (D) 1 2 3 13 14 15 25 26 27

Question 15 of 34 | GATE 2022 Single Session | Q.44 (Computer Science & IT)

Q.44

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
char a = 'P';
char b = 'x';
char c = (a & b) + '*';
char d = (a | b) - '-';
char e = (a ^ b) + '+';
printf("%c %c %c\n", c, d, e);
return 0;
}
```

ASCII encoding for relevant characters is given below

- (A) z K S
- (B) 122 75 83
- (C) * - +
- (D) P x +

Question 16 of 34 | GATE 2022 Single Session | Q.62 (Computer Science & IT)

Q.62

Consider the queues

1

Q containing four elements and

2

Q containing none (shown

as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q,element) and Dequeue(Q). The minimum number of

Enqueue operations on

1

Q required to place the elements of

1

Q in

2

Q in reverse

order (shown as the Final State in the figure) without using any additional storage is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 17 of 34 | GATE 2019 Single Session | Q.24 (Computer Science & IT)

Q.24 Consider the following C program:

```
#include <stdio.h>

int main(){
int arr[]={1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4;
printf("%d\n", ip[1]);
return 0;
}
```

The number that will be displayed on execution of the program is _____ .

CS

6/15

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 34 | GATE 2019 Single Session | Q.40 (Computer Science & IT)

Q.40 Consider the following statements:

I.

The smallest element in a max-heap is always at a leaf node

II.

The second largest element in a max-heap is always a child of the root node

III.

A max-heap can be constructed from a binary search tree in $\Theta(n)$ time

IV.

A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

(A) I, II and III

(B) I, II and IV

(C) I, III and IV

(D) II, III and IV

Question 19 of 34 | GATE 2019 Single Session | Q.52 (Computer Science & IT)

Q.52 Consider the following C program:

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
j = j + j;
sum = sum + i/j;
printf("%f\n", sum);
}
return 0;
}
```

The number of times the variable sum will be printed, when the above program is executed, is _____.

CS

13/15

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 34 | GATE 2019 Single Session | Q.53 (Computer Science & IT)

Q.53

Consider the following C program:

```
#include <stdio.h>
int main()
{
int a[] = {2, 4, 6, 8, 10};
int i, sum = 0, *b = a + 4;
for (i = 0; i < 5; i++)
sum = sum + (*b - i) - *(b - i);
printf ("%d\n", sum);
return 0;
}
```

The output of the above C program is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 21 of 34 | GATE 2018 Single Session | Q.2 (Computer Science & IT)

Q.2

Consider the following C program.

```
#include<stdio.h>
struct Ournode{
char x,y,z;
};
int main(){
struct Ournode p = {'1', '0', 'a'+2};
```

```

struct Ournode *q = &p;
printf ("%c, %c", *((char*)q+1), *((char*)q+2));
return 0;
}

```

The output of this program is:

- (A) 0, c
- (B) 0, a+2
- (C) '0', 'a+2'
- (D) '0', 'c'

Question 22 of 34 | GATE 2018 Single Session | Q.3 (Computer Science & IT)

Q.3

A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.

Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?

- (A) $\theta(1)$, $\theta(1)$
- (B) $\theta(1)$, $\theta(n)$
- (C) $\theta(n)$, $\theta(1)$
- (D) $\theta(n)$, $\theta(n)$ head tail CS 2/20

Question 23 of 34 | GATE 2018 Single Session | Q.29 (Computer Science & IT)

Q.29 Consider the following C program:

```

#include<stdio.h>
void fun1(char *s1, char *s2){
    char *tmp;
    tmp = s1;
    s1 = s2;
    s2 = tmp;
}
void fun2(char **s1, char **s2){
    char *tmp;
    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}

```

```
int main(){
char *str1 = "Hi", *str2 = "Bye";
fun1(str1, str2); printf("%s %s ", str1, str2);
fun2(&str1, &str2); printf("%s %s", str1, str2);
return 0;
}
```

The output of the program above is

- (A) Hi Bye Bye Hi
- (B) Hi Bye Hi Bye
- (C) Bye Hi Hi Bye
- (D) Bye Hi Bye Hi

Question 24 of 34 | GATE 2016 Set 1 (CS1) | Q.10 (Computer Science & IT)

Q.10

A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?

- (A) Both operations can be performed in $O(1)$ time
- (B) At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$
- (C) The worst case time complexity for both operations will be $\Omega(n)$
- (D) Worst case time complexity for both operations will be $\Omega(\log n)$ CS(Set A) 2/17 GATE 2016

Question 25 of 34 | GATE 2016 Set 1 (CS1) | Q.12 (Computer Science & IT)

Q.12

Consider the following C program.

```
void f(int, short);
void main()
{
int i = 100;
short s = 12;
short *p = &s;
_____ ;
// call to f()
}
```

Which one of the following expressions, when placed in the blank above, will NOT result in a type checking error?

- (A) $f(s, *s)$

- (B) $i = f(i, s)$
- (C) $f(i, *s)$
- (D) $f(i, *p)$ CS(Set A) 3/17 GATE 2016

Question 26 of 34 | GATE 2016 Set 1 (CS1) | Q.15 (Computer Science & IT)

Q.15

Consider the following C program.

```
#include<stdio.h>
void mystery(int *ptr a, int *ptr b) {
    int *temp;
    temp = ptr b;
    ptr b = ptr a;
    ptr a = temp;
}
int main() {
    int a=2016, b=0, c=4, d=42;
    mystery(&a, &b);
    if (a < c)
        mystery(&c, &a);
    mystery(&a, &d);
    printf("%d\n", a);
}
```

The output of the program is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 27 of 34 | GATE 2016 Set 1 (CS1) | Q.20 (Computer Science & IT)

Q.20

Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system.

Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

- (A) Shortest remaining time first
- (B) Round-robin with time quantum less than the shortest CPU burst
- (C) Uniform random
- (D) Highest priority first with priority proportional to CPU burst length

Question 28 of 34 | GATE 2016 Set 1 (CS1) | Q.35 (Computer Science & IT)

Q.35

What will be the output of the following C program?

```
void count(int n){
    static int d=1;
    printf("%d ", n);
    printf("%d ", d);
    d++;
    if(n>1) count(n-1);
    printf("%d ", d);
}

void main(){
    count(3);
}
```

- (A) 3 1 2 2 1 3 4 4 4
- (B) 3 1 2 1 1 1 2 2 2
- (C) 3 1 2 2 1 3 4
- (D) 3 1 2 1 1 1 2 CS(Set A) 11/17 GATE 2016

Question 29 of 34 | GATE 2016 Set 1 (CS1) | Q.36 (Computer Science & IT)

Q.36

What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed?

```
a=3;
void n(x) {x = x * a; print(x);}
void m(y) {a = 1; a = y - a; n(a); print(a);}
void main() {m(a);}
```

- (A) 6, 2
- (B) 6, 6
- (C) 4, 2
- (D) 4, 4

Question 30 of 34 | GATE 2016 Set 1 (CS1) | Q.37 (Computer Science & IT)

Q.37

An operator delete(*i*) for a binary heap data structure is to be designed to delete the item in the *i*-th node. Assume that the heap is implemented in an array and *i* refers to the *i*-th index of the array. If the heap tree has depth *d* (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?

- (A) $O(1)$
- (B) $O(d)$ but not $O(1)$
- (C) $O(2d)$ but not $O(d)$
- (D) $O(d \cdot 2d)$ but not $O(2d)$

Question 31 of 34 | GATE 2016 Set 2 (CS2) | Q.14 (Computer Science & IT)

Q.14

The Floyd-Warshall algorithm for all-pair shortest paths computation is based on

- (A) Greedy paradigm.
- (B) Divide-and-Conquer paradigm.
- (C) Dynamic Programming paradigm.
- (D) neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.

Question 32 of 34 | GATE 2016 Set 2 (CS2) | Q.15 (Computer Science & IT)

Q.15

N items are stored in a sorted doubly linked list. For a delete operation, a pointer is provided to the record to be deleted. For a decrease-key operation, a pointer is provided to the record on which the operation is to be performed.

An algorithm performs the following operations on the list in this order: $\Theta(N)$ delete, $O(\log N)$ insert, $O(\log N)$ find, and $\Theta(N)$ decrease-key.

What is the time complexity of all these operations put together?

- (A) $O(\log^2 N)$
- (B) $O(N)$
- (C) $O(N^2)$
- (D) $\Theta(N^2 \log N)$

Question 33 of 34 | GATE 2016 Set 2 (CS2) | Q.34 (Computer Science & IT)

Q.34

A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once.

The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 34 | GATE 2016 Set 2 (CS2) | Q.41 (Computer Science & IT)

Q.41

In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u , and $[u]$ in the adjacency list of v . These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

- (A) $\Theta(n^2)$
- (B) $\Theta(n+m)$
- (C) $\Theta(m^2)$
- (D) $\Theta(n^4)$